FF-S-00109(COM-NBS)
February 21, 1956

INTERIM FEDERAL SPECIFICATION
FOR
SCREWS, WOOD; CROSS-RECESSED HEAD

(This Interim Federal Specification has been developed by the Engineering Metrology Section, National Bureau of Standards, based upon currently available technical information, but has not been approved for promulgation as a regular Federal Specification. It is subject to modification, and, pending its promulgation as a regular coordinated Federal Specification, may be used in procurement.)

1. SCOPE AND CLASSIFICATION

1.1 Scope. -This specification covers wood screws with heads cross-recessed for driving.

1.2 Classification. -Cross-recessed wood screws shall be of the following styles, cross-recess designs, and threads, as specified.

Styles:

Style 1c - Round head
Style 2c - Flat (82° countersunk) head
Style 6c - Oval (82° countersunk) head

Cross-Recess Designs:

Design I
Design II

Threads:

Single thread
Double thread (design I only).

1.3 Sizes. -Cross-recessed head wood screws shall be of the nominal size numbers, basic and shank diameters, and threads per inch as listed in table I.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 The following Federal and Military Specifications and Standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

NN-B-591 - Boxes; Fiberboard, Wood-Cleated (for Domestic Shipments).
NN-B-621 - Boxes; Wood, Nailed and Lock-Corner.
NN-K-231 - Kegs, Wood, Slack.
QQ-A-267 - Aluminum Alloy 2A5, Bars, Rods, and Shapes-Extruded.

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QQ-A-351 - Aluminum Alloy-17S, Bars, Rods, and Wire (Rolled or Drawn.)
QQ-C-320 - Chromium, Plating (Electrodeposited).
QQ-C-591 - Copper-Silicon-Alloy; Bars, Plates, Rods, Shapes, Sheets and Strips.
QQ-M-151 - Metals; General Specification for Inspection of.
QQ-N-290 - Nickel Plating (Electrodeposited).
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-W-321 - Wire, Brass.
QQ-W-396 - Wire, Nickel-Copper-Alloy.
QQ-W-405 - Wire, Steel (Alloy), for Cold Heading and Cold Forging.
QQ-W-409 - Wire, Steel (Carbon), for Cold Heading and Cold Forging.
QQ-W-423 - Wire, Steel, Corrosion-Resisting.
QQ-Z-325 - Zinc Plating (Electrodeposited).
LLL-B -631 - Boxes; Fiber, Corrugated, (for Domestic Shipment).
LLL-B-636 - Boxes; Fiber, Solid, (for Domestic Shipment).
PPP-B-601 - Boxes; Wood, Cleated-Plywood.
PPP-B-676 - Boxes, Set-up, Paper Board.

Military Specifications:

MIL-L-10547 - Liners, Case, Waterproof.
JAN-P-103 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Solid Fiberboard.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed (For Weight of Contents not in excess of 1000 pounds).
JAN-P-108 - Packaging and Packing for Overseas Shipment - Boxes; Fiberboard Corrugated and Solid (Military Type).
MIL-P-116 - Preservation, Methods of.
JAN-P-120 - Packaging and Packing for Overseas Shipment - Cartons, Folding, Paperboard.
MIL-P-3682 - Packaging of Hardware, Builders and Pole Line.
MIL-P-11666 - Packaging of Standard Hardware (for Ordnance Corps use).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. -Wood screws with cross-recessed heads shall be made of aluminum, brass, copper-silicon-alloy, nickel-copper-alloy, carbon steel, alloy steel, or corrosion-resisting steel in accordance with pertinent specifications listed in 2.1 and referenced in the invitation for bids, or of such other material as specified in the invitation for bids.

3.2 Dimensional requirements.

3.2.1 Basic diameters.

3.2.1.1 Single-thread screws. -The basic diameter in single-thread wood screws is the diameter of the unthreaded portion of the screw shank and for each size of screw this basic diameter shall conform to the figures given in table I with permissible tolerances as stated in par. 3.2.1.3.

3.2.1.2 Double-thread screws (design I only). The basic diameter in double-thread wood screws is the outside (major) diameter of the thread and for each size of screw this basic diameter shall conform to the figures given in table I with permissible tolerances as stated in par. 3.2.1.3.

3.2.1.3 Tolerances on basic diameters. -Basic diameters in single- and double-thread wood screws may vary within the tolerances of plue 0.004 and minus 0.007 inch.

3.2.2 Shank diameters. -Shank (unthreaded portion) diameters in single-thread wood screws, being identical with the basic diameters of the screws, shall conform to the requirements in pars. 3.2.1.1 and 3.2.1.3. Shank diameters in double-thread wood screws shall conform to the dimensions given in table I with permissible tolerances: plus 0, minus 0.002 inch.

3.2.3 Heads.

3.2.3.1 Round head wood screws, style 1c, fig. 1. -Head and recess dimensions shall be as given in table II.

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3.2.3.2 Flat head (82°countersunk) wood screws, style 2c, fig. 2. - Head and recess dimensions shall be as given in table III.

3.2.3.3 Oval head (82° countersunk) wood screws, style 6c, fig. 3.- Head and recess dimensions shall be as specified in table IV.

3.2.3.4 Countersink. -The included angle of the countersink of the head in flat-head (style 2c) and oval-head (style 6c) cross-recessed-head wood screws shall be 82 degrees with a permissible variation of plus 0 degrees and minus 2 degrees. (See figs. 2 and 3.)

3.2.3.5 Radius of fillet. -The fillet under the head of style 1c round head wood screws shall have a nominal radius of 1/64 inch for sizes through No. 9, and 1/32 inch for sizes over No. 9.

3.2.3.6 Cross-recesses (figs. 1, 2, 3, 4 and 6).

3.2.3.6.1 Cross-recess, design I. -This design of the cross-recess shall have a large center opening, tapered wings, and a blunt bottom, with all edges relieved or rounded.

3.2.3.6.2 Cross-recess, design II. -This design of the cross-recess shall consist of two intersecting slots with parallel sides converging to a sharp apex at the bottom of the recess.

3.2.3.6.3 Recess dimensions. -Recess dimensions of either design shall assure the penetration, Q, of standard gage points, as specified in tables II, III, and IV. Recesses and gage points shall conform to the recess and gage dimensions, respectively, as given in tables V and VIII and figures 4 and 6.

3.2.3.6.4 Concentricity. -The cross-recesses in all cross-recessed wood screws shall be concentric with the axis of the screw body within a tolerance of 3 percent of the maximum head diameter or 0.010 inch, whichever is greater.

3.2.4 Length.

3.2.4.1 The length of wood screws shall be measured from the largest diameter of the bearing surface of the head to the extreme point in a line parallel to the axis of the screw.

3.2.4.2 Tolerances on length. -Unless otherwise specified, the permissible variations in length of cross-recessed head wood screws, styles 1c, 2c and 6c, shall be as specified in table XI. Unless otherwise specified, the lengths of recessed head wood screws on orders from the Army, Navy, or Air Force shall be within the tolerances given in tables XII and XIII.

3.2.5 Threads.

3.2.5.1 Single-thread wood screws.

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3.2.5.1.1 Length of thread. -The length of the threaded portion shall be equal to approximately two-thirds of the length of the screw.

3.2.5.1.2 Threads per inch. -The number of threads per inch for the various sizes of wood screws shall be as specified in table I. The number of threads per inch may vary within a tolerance of plus or minus 10 percent.

3.2.5.2 Double-thread wood screws (design I only).

3.2.5.2.1 Length of thread. -The length of thread shall be as specified in table XIV.

3.2.5.2.2 Threads per inch. -The number of threads per inch shall be as specified in table I.

3.2.6 Points. -Cross-recessed head wood screws shall be supplied with gimlet, cone, or diamond points at the manufacturer's option unless the shape of the point is specified.

3.3 Finish. -Unless otherwise specified, all cross-recessed head wood screws shall be furnished plain, uncoated, with a naturally bright finish and not heat treated.

3.3.1 When finish is specified, it shall be in accordance with the pertinent specifications listed under 2.1 as referenced in the invitation for bids, constrct, or order.

3.3.2 Conservation. -For the conservation of critical materials under emergency conditions it is recommended the phosphate coating shall be substituted for plating by critical materials, when applicable.

3.4 Workmanship. -All cross-recessed head wood screws shall be of uniform quality and appearance and shall be reasonably free from injurious defects such as rust, scale, fins, seams, cracks and other defects which may affect their use, handling, and serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot. -A lot shall consist of all wood screws of the same style, recess design, length, size, thread, and material, produced by the same manufacturer, under essentially the same conditions, under the same contract or order, and submitted for inspection at the same time.

4.2 Inspection.

4.2.1 Sampling for visual, material, dimensional, and workmanship inspection. -Unless otherwise specified, the Government inspector shall arrange to select at random from each lot a sufficient number of wood screws to satisfy himself as to whether the material conforms to the requirements of this specification.

4.2.1.1 When specified, the sample may be selected on the basis of the approved quality level of approximately 4 percent defectives, as specified in table XV.

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4.2.2 Visual, material, dimensions, finish, and workmanship inspection.

4.2.2.1 When sampling is done as specified in 4.2.1, the Government inspector shall measure and inspect each screw in the sample for material, dimensional, finish, and workmanship requirements and reject the lot if two or more samples do not conform with the requirements of this specification.

4.2.2.2 When sampling is made on basis of the accepted quality level, (see 4.2.1.1), each sample screw shall be measured and visually inspected by the Government inspector for compliance with the specified material, dimensional, finish, and workmanship requirements. If the number of defective screws in the sample is less than or equal to the acceptance number in table XV for the size of the lot under inspection, the lot represented by the sample shall be accepted insofar as the material, dimensional, finish, and workmanship requirements are concerned. If the number of defective screws is equal to or greater than the rejection number shown in table XV for the lot size under inspection, the lot represented by the sample shall be rejected.

4.2.3 Cross-recesses. -Cross-recesses in the screws shall be inspected by the Government inspector for penetration by using the gages shown in figs. 4, 5, 6 and 7 and of specifications given in Tables V, VI, VII, VIII, IX, and X.

4.2.3.1 Inspection gages. -Unless otherwise specified in the contract or order, the manufacturer shall make available to the Government inspector the above specified gages, the accuracy of which shall be certified in a manner satisfactory to the Government inspector. In case of controversy, the certification of gages by the National Bureau of Standards shall govern.

4.2.4 Reinspection. -A lot rejected for failure to meet the constructional, material, dimensional, finish, and workmanship requirements may be resubmitted for Governmental inspection provided that the manufacturer or contractor, after having been notified of the reasons for rejection, has inspected all material in the lot for the deficiency(ies) noted and has removed and replaced all nonconforming material. The Government inspector shall again select at random a new sample as specified in 4.2.1 or 4.2.1.1 and examine it for compliance with this specification. One reinspection only shall be permitted.

4.2.5 Tests.

4.2.5.1 Tests of material and of coating. -Tests of the material or of the coating in the wood screws submitted for inspection are not required unless such tests are specified in the invitation for bids, contract, or order, or are formally requested by the Government inspector. The required tests shall conform to those stated in the pertinent specifications listed under 2.1, or in Federal Specification QQ-M-151.

4.2.5.2 Rejection and retest.

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4.2.5.2.1 Rejection. -If any sample fails to meet the specified test requirements, the entire lot represented by it shall be rejected.

4.2.5.2.2 Retest. -One retest, if requested by the contractor, shall be permitted in accordance with the stipulations in Federal Specification QQ-M-151.

5. PREPARATION FOR DELIVERY.

5.1 The purchasers should specify whether the subject commodity is to be packed and shipping containers marked for domestic or overseas shipment.

5.1.1 The preservation, packaging, packing, and marking requirements specified herein apply only to direct purchases and shipments to the Government.

5.2 Packaging. -Unless otherwise specified, wood screws shall be packaged in units of 1 gross, or fraction thereof, or of 100, or fraction thereof, in standard commercial containers in accordance with good commercial practice.

5.2.1 When specified, wood screws shall be cleaned, preserved, and packaged in accordance with specification MIL-P-3982 (for Ordnance Corps - MIL-P-11666). Unit and intermediate containers shall conform to specifications JAN-P-108, JAN-P-120, JAN-P-133, PPP-B-676, LLL-B-631, LLL-B-636 respectively. Slack kegs shall conform to specification NN-K-231. Closure, sealing, and the gross weight of the containers shall conform to the requirements in the applicable container specifications. The gross weight of slack kegs shall not exceed approximately 200 pounds.

5.3 Packing.

5.3.1 Packing for domestic shipment (immediate use). -Unless otherwise specified, the packaged wood screws shall be packed in standard commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to afford maximum protection against normal hazards of transportation and assure acceptance by the carriers and safe delivery to the designated point at the lowest rate of the carrier.

5.3.2 For domestic shipment and storage. -Unless otherwise specified, the packaged wood screws, except those packaged in slack kegs, shall be packed in boxes conforming to specifications NN-B-591, NN-B-621, LLL-B-631, LLL-B-636 or PPP-B-601, respectively. The gross weight of wood boxes shall not exceed 200 pounds and that of fiber boxes, 70 pounds. Slack kegs, where used, will require no overpacking.

5.3.3 For overseas shipment. -When specified, the packaged wood screws, except those packaged in slack kegs, shall be packed in boxes conforming to specifications JAN-P-103, JAN-P-106 (style 2 or 4), JAN-P-108 (symbol V3c or V3s), respectively. Boxes shall be lined with a waterproof case liner conforming to grade A or B of specification MIL-L-10547 and sealed as specified therein. Shipping containers shall be sealed and strapped as specified in the pertinent specifications. The gross weight of wood boxes shall

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not exceed 150 pounds; that of fiber-board boxes, 70 pounds, and that of nailed slack kegs, 200 pounds. Case liners will not be required in fiber boxes conforming to specification JAN-P-108. Slack kegs shall conform to specification NN-K-231 and will require no overpacking.

5.3.4 Where practicable, all containers shall contain an identical number of items and shall be of uniform size. The containers shall be new and shall be able to withstand storage, rehandling, and reshipment without necessity of repacking.

5.3.5 Pilot pack sampling. -When specified, a pilot pack prior to packing of an item in quantity shall be submitted to the Government inspector to determine conformance with the sampling, inspection, and test procedures of specification MIL-P-116. The submission of a pilot pack will not be required when the prime or subcontractor has previously had such a pack tested and accepted on a military order and can furnish satisfactory evidence that the material is being packed identically with the approved pack. Approval of the pilot pack shall not relieve the contractor of his obligation to preserve, package, and pack the material in conformance with specification MIL-P-116 and this specification.

5.4 Marking.

5.4.1 Issue packages. -Unless otherwise specified, the issue packages shall be marked with the name, length, nominal size number, head style, cross-recess design, thread (single or double), material, manufacturer's name or trade mark, contractor's name, if different from the manufacturer's, and with this Interim Federal Specification symbol.

5.4.2 Shipping containers. -Unless otherwise specified, shipping containers shall be marked or carry oil-proof or metal tags on one end or side with the following information:

Interim Federal Specification Symbol: FF-S-00109 (COM-NBS)

Wood screws, cross-recessed
Length and the numbered size
Material
Head style and recess design
Thread (single or double)
Finish (if specified)
Quantity
Consignee's name
Manufacturer's name or trademark
Contractor's name (if not the same as the manufacturer's)
Gross weight

5.4.3 Military marking. -In addition to any special marking required by the contract or order, unit packages, intermediate, and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be: Screws, Wood, Cross-Recessed Head.

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6. NOTES

6.1 Ordering Data. -Purchasers should specify:

Interim Federal Specification title, symbol, and date
Head style required
Recess design
Thread (single and double)
Material
Length and numbered size
Quantity required
Finish (if required)
Packaging: in cartons or in bulk (bags)
Packing: for domestic (immediate use), domestic (storage) or overseas shipment

and should exercise any desired option offered herein, see pars: 3.1, 3.2.6, 3.3, 3.3.1, 4.2.1, 4.2.1.1, 4.2.3.1, 4.2.5.1, 5.1, 5.2, 5.3, 5.4.1, and 5.4.2.

6.1.1 Wood screws are designated commercially by the length and the numbered size. The length is given first, for example: 1-1/2 inch, No. 12.

6.1.2 Although wood screws can be obtained in any material and size when ordered in sufficient quantities to justify special tooling, the purchases of steel and brass screws should be confined, insofar as practicable, to the sizes shown in tables XVI, XVII and XVIII which list standard stock production sizes and lengths.

6.2 Transportation description. -The applicable transportatation description:

Screws, wood, cross-recessed head, plain or galvanized

Carload, minimum weight, 40,000 pounds
Truckload, minimum weight, 36,000 pounds

6.3 Miscellaneous Notes.

6.3.1 The wood screws with cross-recessed heads of design I are commercially known as "Phillips" recessed-head wood screws and the wood screws with cross-recessed heads of the design II as "Frearson" (or "Reed and Prince") recessed-head wood screws. Both designs of cross recesses assure correct centering and concentration of the driving power towards the screw point.

6.3.2 The dimensional requirements for cross-recesses and for the gages for measuring them specified in this specification are in conformance with the standards published in American Standard ASA B18.6.1-1954, Slotted and Recessed Head Wood Screws.

6.3.3 Style numbers. -Style numbers, listed in this specification, relate to head styles and are in accordance with the uniform designations for various types of machine and wood screws approved for use by all Federal agencies as listed in table XVIII.

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6.4 Federal Specifications do not include all styles, sizes, etc., of the commodities indicated by the titles of the specification which are commercially available, but are intended to cover the types, classes, etc., which are suitable for Federal Government requirements.

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TABLE I.- Standard nominal sizes, basic and shank diameters and threads in cross-recessed-head wood screws

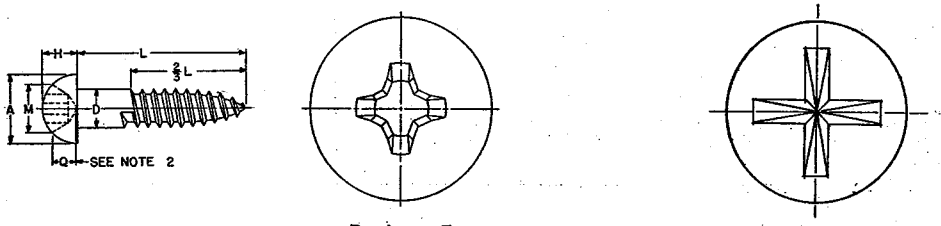
Nominal size number	Single-thread screws		Double-thread screws		
	Basic diameter: shank 1/	Threads per inch	Diameters		Threads per inch
			Basic: O.D. thread 1/	Shank 2/	
	Inch	No.	Inch	Inch	No.
0	0.060	32	———	-----	---
1	.073	28	-----	-----	---
2	.086	26	0.086	0.066	28
3	.099	24	.099	.081	26
4	.112	22	.112	.090	24
5	.125	20	.125	.102	22
6	.138	18	.138	.111	20
7	.151	16	.151	.122	18
8	.164	15	.164	.131	18
9	.177	14	.177	.141	16
10	.190	13	.190	.152	15
12	.216	11	.216	.171	13
14	.242	10	.242	.196	12
16	.268	9	.268	.209	10
18	.294	8	.294	.232	10
20	.320	8	-----	-----	---
24	.372	7	-----	-----	---

1/ Tolerances: Plus 0.004, minus 0.007 inch.

2/ Tolerances: Plus 0.002, minus 0.000

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FIGURE 1. ROUND HEAD, CROSS-RECESSED WOOD SCREWS



Design I

Design II

TABLE II.- Head proportions of round head cross-recessed wood screws, style 1c

NOMINAL SIZE AND DIAMETER		DIMENSIONS OF HEAD[1]						DIMENSIONS OF DESIGN I RECESS						DIMENSIONS OF DESIGN II RECESS					
		A		H		M		T		N				M		T	N		
		Diameter of head		Height of head		Diameter		Depth		Width	Driver size	Depth of penetration[2]		Diameter		Depth	Width	Depth of penetration[2]	
		Maximum	Minimum	Maximum	Minimum	Max.	Min.	Max.	Min.	Min.		Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.
No.	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch		Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch
0	0.060	0.113	0.099	0.053	0.043	0.070	0.060	0.039	0.022	0.014	0	0.032	0.015	0.085	0.074	0.054	0.021	0.024	0.009
1	.073	.138	.122	.061	.051	.079	.069	.049	.033	.015	0	.042	.026	.103	.091	.067	.024	.046	.029
2	.086	.162	.146	.069	.059	.111	.101	.061	.042	.018	1	.054	.035	.121	.109	.079	.027	.064	.048
3	.099	.187	.169	.078	.067	.119	.109	.070	.052	.019	1	.063	.045	.140	.127	.091	.030	.081	.064
4	.112	.211	.193	.086	.075	.127	.117	.080	.063	.018	1	.073	.056	.158	.145	.103	.032	.097	.081
5	.125	.236	.217	.095	.083	.151	.141	.071	.046	.027	2	.060	.035	.177	.163	.115	.035	.114	.096
6	.138	.260	.240	.103	.091	.159	.149	.081	.056	.027	2	.070	.045	.195	.180	.127	.038	.119	.101
7	.151	.285	.264	.111	.099	.167	.157	.089	.066	.029	2	.078	.055	.214	.198	.139	.040	.137	.117
8	.164	.309	.287	.120	.107	.175	.165	.098	.075	.030	2	.087	.064	.232	.215	.151	.043	.153	.132
9	.177	.334	.311	.128	.115	.183	.173	.107	.084	.030	2	.096	.073	.250	.233	.163	.046	.169	.148
10	.190	.359	.334	.137	.123	.192	.182	.116	.093	.031	2	.105	.082	.269	.250	.176	.048	.185	.162
12	.216	.408	.382	.153	.139	.246	.236	.122	.099	.036	3	.105	.082	.306	.286	.200	.054	.194	.170
14	.242	.457	.429	.170	.155	.262	.252	.139	.116	.037	3	.122	.099	.343	.322	.224	.059	.228	.204
16	.268	.506	.476	.187	.171	.278	.268	.156	.136	.039	3	.139	.119	.379	.357	.248	.064	.261	.235
18	.294	.555	.523	.204	.187	.336	.326	.173	.151	.057	4	.153	.131	.416	.392	.270	.070	.293	.266
20	.320	.604	.570	.220	.203	.353	.343	.191	.168	.059	4	.174	.148	.453	.427	.294	.075	.325	.296
24	.372	.702	.664	.254	.235	.384	.374	.225	.202	.064	4	.205	.182	.527	.498	.342	.086	.347	.315

1/ For radius of fillet under head see paragraph 3.2.3.5.

2/ All recess dimensions are controlled by the penetration Q of standard gage points as specified in tables VI, VII, and VIII.

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FIGURE 2. FLAT HEAD, (82° COUNTERSUNK) CROSS - RECESSED WOOD SCREWS

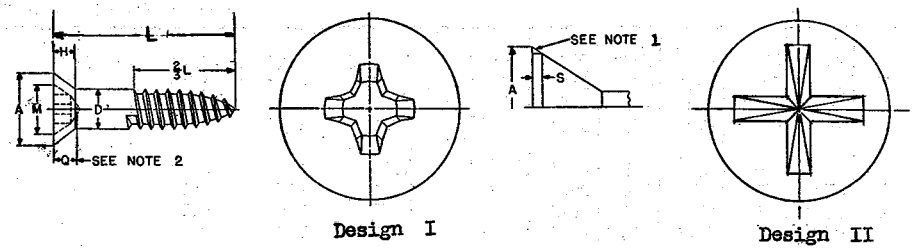


TABLE III.- Head proportions of flat head (82° countersunk) cross-recessed wood screws, style 2c

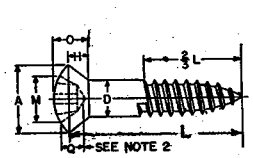
NOMINAL SIZE AND BASIC DIAMETER		DIMENSIONS OF HEAD[1]					DIMENSIONS OF DESIGN I RECESS								DIMENSIONS OF DESIGN II RECESS					
		A			H		M		T		N				M		T	N		
		Diameter of head[1]			Height of head		Diameter		Depth		Width	Driver size	Depth of penetration[2]		Diameter		Depth	Width	Depth of penetration[2]	
		Maximum sharp	Minimum sharp	Absolute minimum with flat	Maximum	Minimum	Max.	Min.	Max.	Min.	Min.		Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.
No.	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch		Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch
0	0.060	0.119	0.105	0.099	0.035	0.026	0.066	0.056	0.040	0.027	0.014	0	0.033	0.020	0.090	0.079	0.059	0.022	0.033	0.022
1	.073	.146	.130	.123	.043	.033	.074	.064	.048	.035	.015	0	.041	.028	.109	.097	.071	.025	.045	.034
2	.086	.172	.156	.147	.051	.040	.099	.089	.060	.047	.017	1	.053	.040	.129	.117	.084	.028	.058	.047
3	.099	.199	.181	.171	.059	.048	.104	.094	.065	.052	.018	1	.058	.045	.149	.136	.097	.031	.071	.060
4	.112	.225	.207	.195	.067	.055	.125	.115	.086	.073	.018	1	.079	.066	.169	.155	.110	.034	.084	.072
5	.125	.252	.232	.220	.075	.062	.151	.141	.083	.063	.027	2	.072	.052	.189	.174	.123	.037	.097	.084
6	.138	.279	.257	.244	.083	.069	.171	.161	.103	.083	.029	2	.092	.072	.209	.193	.136	.039	.110	.097
7	.151	.305	.283	.268	.091	.076	.186	.176	.118	.098	.031	2	.107	.087	.229	.212	.149	.042	.123	.109
8	.164	.332	.308	.292	.100	.084	.201	.191	.133	.113	.033	2	.122	.102	.249	.231	.162	.045	.136	.122
9	.177	.358	.334	.316	.108	.091	.211	.201	.143	.123	.033	2	.132	.112	.269	.250	.175	.048	.149	.134
10	.190	.385	.359	.340	.116	.098	.255	.245	.143	.125	.037	3	.126	.106	.289	.269	.188	.051	.162	.146
12	.216	.438	.410	.389	.132	.112	.280	.270	.168	.148	.040	3	.151	.131	.329	.308	.214	.057	.188	.172
14	.242	.491	.461	.437	.148	.127	.300	.290	.188	.168	.042	3	.171	.151	.368	.346	.240	.063	.214	.196
16	.268	.544	.512	.485	.164	.141	.324	.314	.213	.193	.045	3	.196	.176	.408	.384	.266	.068	.240	.221
18	.294	.597	.563	.534	.180	.155	.375	.365	.227	.208	.063	4	.207	.188	.448	.423	.292	.074	.266	.247
20	.320	.650	.614	.582	.196	.170	.390	.380	.242	.223	.065	4	.222	.203	.488	.460	.318	.080	.292	.271
24	.372	.756	.716	.679	.228	.198	.421	.411	.273	.254	.070	4	.253	.234	.568	.537	.370	.091	.344	.321

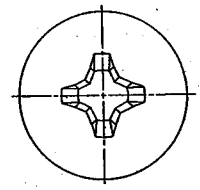
1/ Edges of head may be rounded or straight-sided.

2/ All recess dimensions are controlled by the penetration Q of standard gage points as specified in tables VI, VII, and VIII.

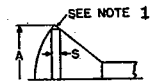
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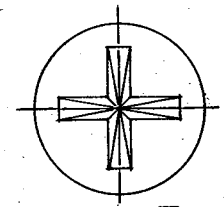
FIGURE 3. OVAL HEAD, (82° COUNTERSUNK) CROSS - RECESSED WOOD SCREWS





Design I





Design II

TABLE IV.- Head proportions of oval head (82° countersunk) cross recessed wood screws, style 6c

NOMINAL SIZE AND DIAMETER		DIMENSIONS OF HEAD							DIMENSIONS OF DESIGN I RECESS							DIMENSIONS OF DESIGN II RECESS					
		A			H		O		M		T		N			M		T	N		
		Diameter of head[1]			Countersink height		Total height of head		Diameter		Depth		Width	Driver size	Depth of penetration[2]	Diameter		Depth	Width	Depth of penetration[2]	
		Maximum sharp	Minimum sharp	Absolute minimum with flat	Maximum	Minimum	Maximum	Minimum	Max.	Min.	Max.	Min.	Min.		Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.
No.	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch		Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch
0	0.060	0.119	0.105	0.099	0.035	0.026	0.056	0.041	0.071	0.061	0.042	0.027	0.012	0	0.035	0.020	0.090	0.079	0.059	0.022	0.032	0.019
1	.073	.146	.130	.123	.043	.033	.068	.052	.074	.064	.045	.030	.015	1	.038	.023	.109	.097	.071	.025	.049	.035
2	.086	.172	.156	.147	.051	.040	.080	.063	.109	.099	.065	.052	.018	1	.059	.045	.129	.117	.084	.028	.066	.052
3	.099	.199	.181	.171	.059	.048	.092	.073	.121	.111	.078	.064	.019	1	.071	.057	.149	.136	.097	.031	.082	.067
4	.112	.225	.207	.195	.067	.055	.104	.084	.133	.123	.091	.077	.019	1	.084	.070	.169	.155	.110	.034	.098	.083
5	.125	.252	.232	.220	.075	.062	.116	.095	.155	.145	.082	.061	.028	2	.071	.050	.189	.174	.123	.037	.107	.091
6	.138	.279	.257	.244	.083	.069	.128	.105	.175	.165	.102	.080	.030	2	.091	.069	.209	.193	.136	.039	.124	.106
7	.151	.305	.283	.268	.091	.076	.140	.116	.186	.176	.112	.092	.031	2	.101	.081	.229	.212	.149	.042	.140	.122
8	.164	.332	.308	.292	.100	.084	.152	.126	.202	.192	.128	.106	.033	2	.117	.095	.249	.231	.162	.045	.156	.137
9	.177	.358	.334	.316	.108	.091	.164	.137	.213	.203	.141	.119	.034	2	.130	.108	.269	.250	.175	.048	.154	.134
10	.190	.385	.359	.340	.116	.098	.176	.148	.258	.248	.139	.118	.037	3	.122	.101	.289	.269	.188	.051	.172	.151
12	.216	.438	.410	.389	.132	.112	.200	.169	.280	.270	.162	.140	.040	3	.145	.123	.329	.308	.214	.057	.205	.184
14	.242	.491	.461	.437	.148	.127	.224	.190	.302	.292	.185	.163	.042	3	.168	.146	.368	.346	.240	.063	.237	.215
16	.268	.544	.512	.485	.164	.141	.248	.212	.329	.319	.211	.189	.046	3	.194	.172	.408	.384	.266	.068	.269	.245
18	.294	.597	.563	.534	.180	.155	.272	.233	.378	.368	.223	.202	.064	4	.203	.182	.448	.423	.292	.074	.301	.276
20	.320	.650	.614	.582	.196	.170	.296	.254	.397	.387	.242	.221	.066	4	.222	.201	.488	.460	.318	.080	.303	.276
24	.372	.756	.716	.679	.228	.198	.344	.297	.433	.423	.279	.257	.072	4	.259	.238	.568	.537	.370	.091	.371	.340

1/ Edges of head may be rounded or straight-sided.

2/ All recess dimensions are controlled by the penetration Q of standard gage points as specified in tables VI, VII, and VIII.

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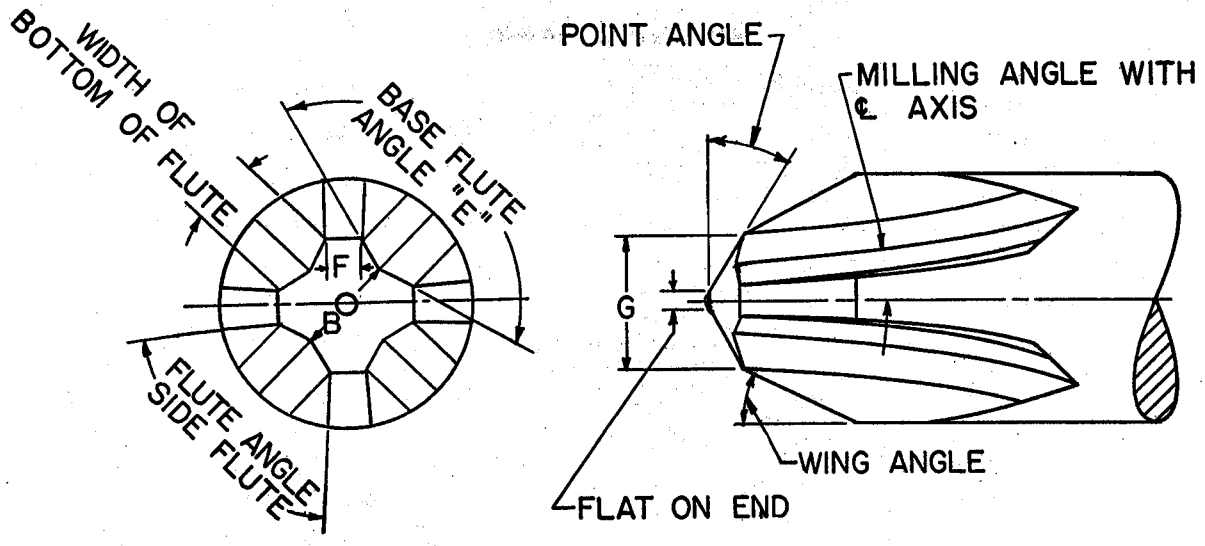


Fig. 4 Design I Cross Recess and Point in Penetration Gage
(See Fig. 5 and Tables V, VI and VII.)

TABLE V.- Dimensions of Design I Cross Recesses, (See Fig. 4)

D I M E N S I O N S	R E C E S S N U M B E R			
	1	2	3	4
G (+.002, -.000)----------Inch	0.050	0.090	0.150	0.200
B (±.001)----------Inch	0.0380 (+.000) (-.001)	0.0578	0.0950	0.1370
Milling angle (+15', -0')----------	7°	5°45'	5°45'	7°
Point angle (+ 0°, - 1°)----------	28°	28°	28°	28°
Wing angle (+ 30', - 0')----------	26°30'	26°30'	26°30'	26°30'
Base flute angle (+0', - 15')----------	138°	140°	146°	153°
Side flute angle (+ 0', - 15')----------	104°	92°	92°	92°
Width of base flute D (+ .000, - .002)-----Inch	0.018	0.033	0.080	0.096
Wing thickness F----------Inch	0.018-0.021	0.025-0.028	0.030-0.033	0.042-0.045

TABLE VI.- Dimensions of Penetration Gage Points for Design I Cross Recesses, (See Fig. 4).

D I M E N S I O N S	R E C E S S N U M B E R			
	1	2	3	4
G (+ .001, - .000)----------Inch	0.050	0.090	0.150	0.200
B (+ .000, - .001)----------Inch	.0394	.0606	.0983	.1407
Point angle (+ 1°, - 0°)----------	18°	18°	18°	18°
Milling angle (+ 0', - 15')----------	7°	5°	5°	7°
Flat on end (+ .000, - .005)----------Inch	.020	.020	.020	.020
Wing angle (+ 0', - 15')----------	26°30'	26°30'	26°30'	26°30'
Angle E (+ 15' - 0')----------	138°	140°	146°	153°
Side flute angle (+ 15', - 0')----------	104°	92°	92°	92°
Wing thickness F (+ .000, - .003)----------Inch	.021	.028	.033	.045
Width of bottom of flute (+ .000, - .001)---Inch	.0202	.0434	.0826	.1078

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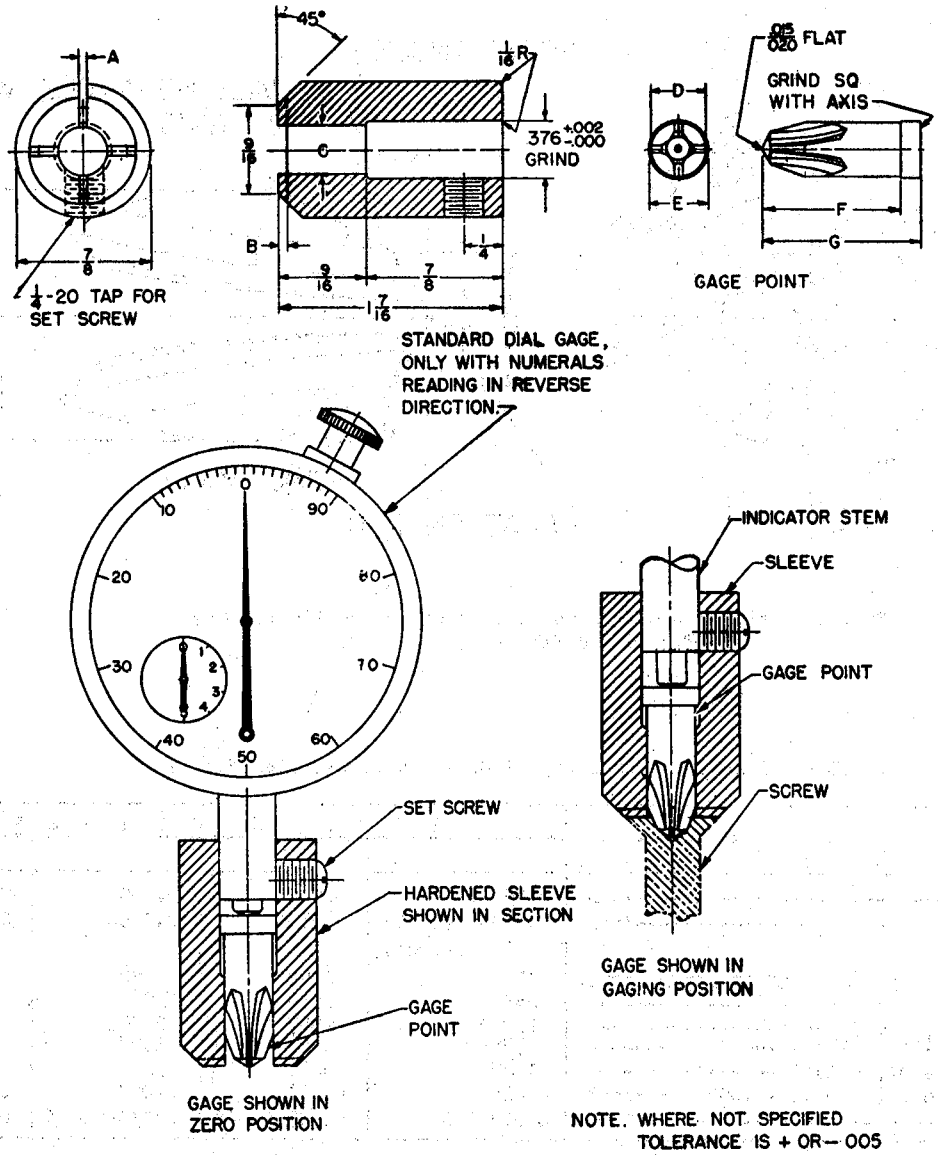


Fig. 5. Penetration gage for design I cross recesses (see fig. 4 and tables VI and VII.)

TABLE VII.- Dimensions of gage sleeves and points in penetration gages for design I cross recesses (See fig. 5)

Size of recess gage	A ±0.002	B ±0.003	C ±0.0001	D ±0.0001	E ±0.005	F ±0.005	G ±0.005
No.	Inch	Inch	Inch	Inch	Inch	Inch	Inch
1.....	0.012	0.020	0.0875	0.0871	0.156	0.688	0.812
2.....	.018	.031	.1417	.1413	.219	.750	.875
3.....	.022	.037	.2097	.2093	.250	.781	.906
4.....	.031	.062	.3126	.3122	.359	.844	.969

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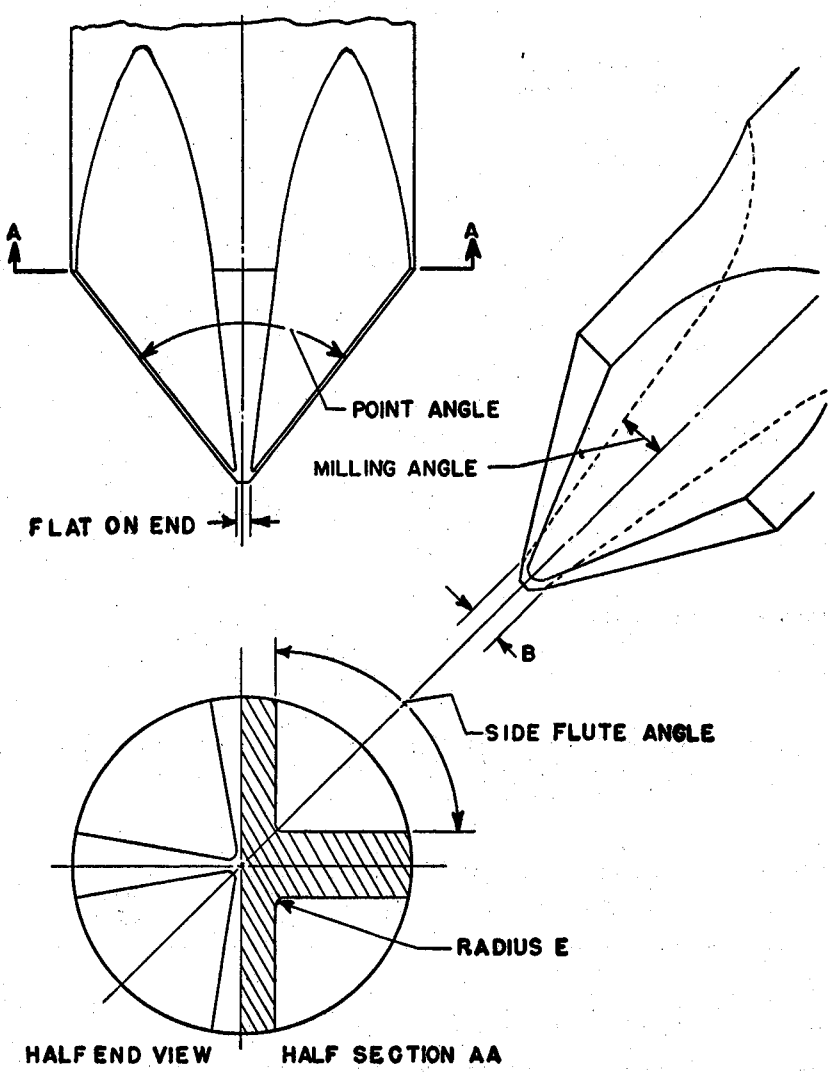


Fig. 6. Design II Cross Recess and Point in Penetration Gage (See Fig. 7 and Tables VII, IX and X.)

TABLE VIII.- Dimensions of Design II Cross Recesses and of Penetration Gage Points (See Fig. 6).

Dimensions	Cross Recess	Penetration Gage Point
B----------Inch	0.0378 (+ 0, - .002)	0.018 (+ .001, - .000)
Milling angle----------	9° (+ 0', - 15')	8°45' (+ 5', - 0')
Point angle----------	75° (+ 1°, - 0°)	75° (+ 0', - 30')
Side flute angle----------	89°17' (± 15')	89°17' (+ 15', - 0')
Radius E----------Inch	0.015 (+ .002, - .000)	0.005 (+ .000, - .001)
Flat on end----------Inch	Sharp Point	0.025 - 0.030

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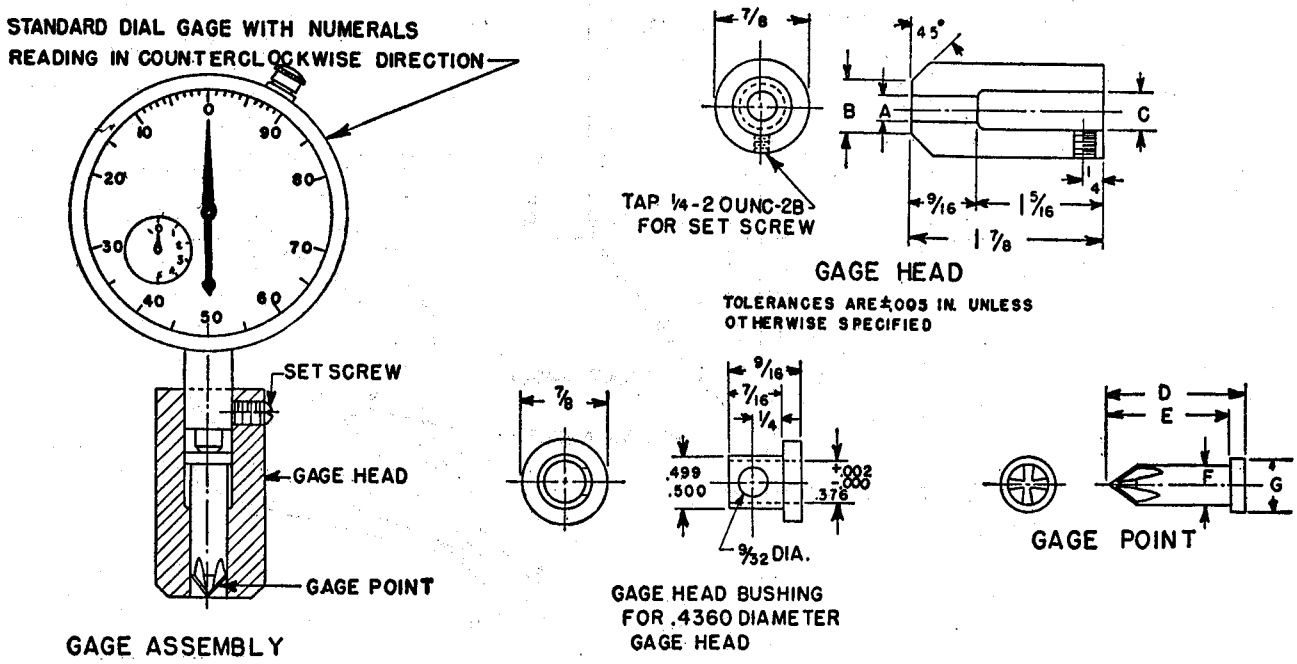


Fig. 7. Penetration Gage for Design II Cross Recesses.
(See Fig. 6 and Tables VIII, IX and X.)

TABLE IX.- Dimensions of Gage Heads and Points in Penetration Gages for Design II Cross Recesses (See Fig. 7).

A[1] ±0.001	B ±0.005	C -0.002 +0.000	D ±0.005	E ±0.005	F ±0.0001	G ±0.005
Inch	Inch	Inch	Inch	Inch	Inch	Inch
0.0930	0.344	0.376	1.000	0.875	0.0926	0.188
.1410	.375	.376	1.000	.875	.1406	.250
.2460	.500	.376	1.062	.938	.2456	.312
.4360	.688	.500	1.250	1.125	.4356	.469

[1] A = gage diameter.

TABLE X.- Styles and Sizes of Cross-Recessed Wood Screws, Design II, and of Gage Diameters in Corresponding Penetration Gages

Style of Screws	GAGE DIAMETER (A)			
	0.093	0.141	0.246	0.436
	Nominal size of wood screws			
1c Round head	2, 3, 4	5, 6, 7, 8, 10	12, 14, 16, 18, 20	24
2c Flat head	2, 3, 4	5, 6, 7, 8	9, 10, 12, 14, 16, 18	20, 24
6c Oval head	2, 3, 4	5, 6, 7, 8	9, 10, 12, 14, 16, 18	20, 24

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TABLE XI. - Tolerances on length in single-thread cross-recessed head wood screws

Length of screws	Tolerance	
	Plus	Minus
Inches	Inch	Inch
1/4 to 5/8, inclusive	0	1/32
3/4 to 1-3/8, inclusive	0	3/64
1-1/2 to 2-3/4, inclusive	0	1/16
3 and over, inclusive	0	3/32

TABLE XII. - Tolerances on length in design I cross-recessed round head, single thread, wood screws, (on orders from Army, Navy, and Air Force)

Screw No.	Tolerance		Screw No.	Tolerance	
	Plus	Minus		Plus	Minus
	Inch	Inch		Inch	Inch
2	0	0.08	10	0	0.13
3	0	.08	12	0	.15
4	0	.09	14	0	.16
5	0	.10	16	0	.18
6	0	.10	18	0	.20
7	0	.11	20	0	.22
8	0	.12	24	0	.27
9	0	.13			

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TABLE XIII. - Limits on length of design I cross-recessed flat and oval head, single-thread, wood screws, (on orders from Army, Navy and Air Force)

Nominal length	Maximum length	Minimum length	Nominal length	Maximum length	Minimum length
Inch	Inch	Inch	Inch	Inch	Inch
1/4	0.250	0.22	2	2.000	1.94
3/8	.375	.34	2-1/4	2.250	2.19
1/2	.500	.46	2-1/2	2.500	2.43
5/8	.625	.59	2-3/4	2.750	2.68
3/4	.750	.71	3	3.000	2.92
7/8	.875	.83	3-1/2	3.500	3.42
1	1.000	.96	4	4.000	3.91
1-1/4	1.250	1.20	4-1/2	4.500	4.40
1-1/2	1.500	1.45	5	5.000	4.89
1-3/4	1.750	1.69	-----	-----	-----

TABLE XIV. - Length of thread of double-thread cross-recessed wood screws

Screw length	Thread length	Screw length	Thread length	Screw length	Thread length
Inch	Inch	Inches	Inches	Inches	Inches
1/4	To head	7/8	9/16	2	1-1/8
3/8	To head	1	5/8	2-1/4	1-1/2
1/2	3/8	1-1/4	3/4	2-1/2	1-1/2
5/8	3/8	1-1/2	15/16	2-3/4	1-3/4
3/4	1/2	1-3/4	1-1/8	3	1-3/4

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TABLE XV. - Sampling for visual, material, finish, dimensional, and workmanship inspection under acceptable quality level (AQL) approximately 4% defectives

Number of screws in inspection lot	Number of screws in sample	Acceptance number (defectives)	Rejection number (defectives)
110 or under	10	1	2
111 - 300	15	1	2
301 - 500	25	2	3
501 - 800	35	3	4
800 - 3,200	50	4	5
3,201 - 8,000	75	6	7
8,000 - 22,000	110	8	9
22,000 - 110,000	150	11	12
110,000 and over	300	20	21

Note: The above table specifies a single sampling procedure. Alternate double or multiple sampling procedures or alternate acceptable quality levels, especially for inspection during continuous production of screws or for different classes of defects, may be specified at the discretion of the procuring agency in accordance with the procedures specified in MIL-STD-105.

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TABLE XVI. - Stock production sizes of cross-recessed head, design I, single thread, STEEL WOOD SCREWS
(F = flat head; R = round head; O = oval head)

Length	No. 2 dia. 0.086 inch	No. 3 dia. 0.099 inch	No. 4 dia. 0.112 inch	No. 5 dia. 0.125 inch	No. 6 dia. 0.138 inch	No. 7 dia. 0.151 inch	No. 8 dia. 0.164 inch	No. 9 dia. 0.177 inch	No. 10 dia. 0.190 inch	No. 12 dia. 0.216 inch	No. 14 dia. 0.242 inch	No. 16 dia. 0.268 inch	No. 18 dia. 0.294 inch
Inches													
1/4	FR	-----	FR	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
3/8	FR	FR	FRO	FR	FRO	-----	-----	-----	-----	-----	-----	-----	-----
1/2	FR	FR	FRO	FRO	FRO	FR	FR	-----	-----	-----	-----	-----	-----
5/8	FR	FR	FRO	FRO	FRO	FR	FRO	-----	-----	-----	-----	-----	-----
3/4	-----	-----	FRO	FRO	FRO	FRO	FRO	FR	FR	FR	-----	-----	-----
7/8	-----	-----	FR	FR	FRO	FR	FRO	FR	FR	FR	-----	-----	-----
1	-----	-----	FRO	FRO	FRO	FRO	FRO	FR	FRO	FR	FR	F	-----
1-1/4	-----	-----	-----	-----	FRO	FRO	FRO	FR	FRO	FRO	FR	F	-----
1-1/2	-----	-----	-----	-----	FRO	FRO	FRO	FR	FRO	FRO	FR	F	F
1-3/4	-----	-----	-----	-----	-----	-----	FR	F	FRO	FRO	FR	F	F
2	-----	-----	-----	-----	-----	-----	FR	FR	FRO	FRO	FR	F	F
2-1/4	-----	-----	-----	-----	-----	-----	-----	-----	F	F	F	-----	-----
2-1/2	-----	-----	-----	-----	-----	-----	FR	F	FR	FR	FR	F	F
3	-----	-----	-----	-----	-----	-----	-----	-----	FR	FR	FR	F	F
3-1/2	-----	-----	-----	-----	-----	-----	-----	-----	F	F	F	F	F
4	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	F	F	F

NOTE: Screws of heads, sizes, and lengths not regularly stocked may be available on orders of sufficient quantities.

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TABLE XVII. - Stock production sizes of cross-recessed head design I, single-thread
BRASS WOOD SCREWS
(F = flat head; R = round head; O = oval head)

Length	No. 2 dia. 0.086 inch	No. 3 dia. 0.099 inch	No. 4 dia. 0.112 inch	No. 5 dia. 0.125 inch	No. 6 dia. 0.138 inch	No. 7 dia. 0.151 inch	No. 8 dia. 0.164 inch	No. 9 dia. 0.177 inch	No. 10 dia. 0.190 inch	No. 12 dia. 0.216 inch	No. 14 dia. 0.242 inch
Inches											
1/4	FR	-----	FR	-----	-----	-----	-----	-----	-----	-----	-----
3/8	FR	F	FRO	-----	F	-----	-----	-----	-----	-----	-----
1/2	FR	F	FRO	FRO	FRO	-----	FR	-----	-----	-----	-----
5/8	-----	-----	FRO	FRO	FRO	-----	FRO	-----	-----	-----	-----
3/4	-----	-----	FRO	FRO	FRO	FR	FRO	FR	FR	-----	-----
7/8	-----	-----	-----	F	FRO	F	FRO	F	FR	-----	-----
1----	-----	-----	-----	FRO	FRO	FRO	FRO	FR	FRO	FR	-----
1-1/4	-----	-----	-----	-----	FRO	FRO	FRO	FR	FRO	FR	F
1-1/2	-----	-----	-----	-----	FRO	FR	FRO	FR	FRO	FR	F
1-3/4	-----	-----	-----	-----	-----	-----	FR	-----	FR	FR	F
2----	-----	-----	-----	-----	-----	-----	FRO	-----	FRO	FR	FR
2-1/2	-----	-----	-----	-----	-----	-----	-----	-----	FR	FR	FR
3----	-----	-----	-----	-----	-----	-----	-----	-----	-----	F	F

NOTE: Screws of head, sizes, and lengths not regularly stocked may be available on orders of sufficient quantities.

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TABLE XVIII. - Stock production sizes of crsserecessed, design I, double-thread, ROUND, FLAT, AND OVAL HEAD WOOD SCREWS
(S = steel screws; B = brass screws)

Length	No. 2 dia. 0.086 inch		No. 3 dia. 0.099 inch		No. 4 dia. 0.112 inch		No. 5 dia. 0.125 inch		No. 6 dia. 0.138 inch		No. 7 dia. 0.151 inch		No. 8 dia. 0.164 inch		No. 9 dia. 0.177 inch		No. 10 dia. 0.190 inch		No. 12 dia. 0.216 inch		No. 14 dia. 0.242 inch		No. 16 dia. 0.268 inch		No. 18 dia. 0.294 inch	
Inches																										
1/4	S	B	-	-	S	B	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3/8	S	B	S	B	S	B	S	-	S	B	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1/2	S	B	S	B	S	B	S	B	S	B	S	-	S	B	-	-	-	-	-	-	-	-	-	-	-	-
5/8	S	-	S	-	S	B	S	B	S	B	S	-	S	B	-	-	-	-	-	-	-	-	-	-	-	-
3/4	-	-	-	-	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	-	-	-	-	-	-	-
7/8	-	-	-	-	S	-	S	B	S	B	S	B	S	B	S	B	S	B	S	-	-	-	-	-	-	-
1----	-	-	-	-	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	-	S	-	-	-
1-1/4	-	-	-	-	-	-	-	-	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	-	-	-
1-1/2	-	-	-	-	-	-	-	-	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	-	S	-
1-3/4	-	-	-	-	-	-	-	-	-	-	-	-	S	B	S	-	S	B	S	B	S	B	S	-	-	-
2----	-	-	-	-	-	-	-	-	-	-	-	-	S	B	S	-	S	B	S	B	S	B	S	-	S	-
2-1/4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	S	-	S	-	S	-	-	-	-	-
2-1/2	-	-	-	-	-	-	-	-	S	-	S	-	S	-	S	-	S	B	S	B	S	B	S	-	S	-
3-----	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	S	-	S	B	S	B	S	-	S	-

NOTE: Screws of sizes and lengths not regularly stocked may be available on orders of sufficient quantities.

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TABLE XIX. - Stock production sizes of cross-recessed, design II, single-threaded
ROUND, FLAT, AND OVAL HEAD WOOD SCREWS
(S = steel screws; B = brass and phosphor bronze screws)

Length	No. 2 dia. 0.086 inch		No. 3 dia. 0.099 inch		No. 4 dia. 0.112 inch		No. 5 dia. 0.125 inch		No. 6 dia. 0.138 inch		No. 7 dia. 0.151 inch		No. 8 dia. 0.164 inch		No. 9 dia. 0.177 inch		No. 10 dia. 0.190 inch		No. 12 dia. 0.216 inch		No. 14 dia. 0.242 inch		No. 16 dia. 0.268 inch		No. 18 dia. 0.294 inch		No. 20 dia. 0.320 inch		No. 24 dia. 0.372 inch	
Inches																														
1/4	S	B	S	B	S	B	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3/8	S	B	S	B	S	B	S	B	S	B	S	-	S	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1/2	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	-	S	-	-	-	-	-	-	-	-	-	-	-	-	-
5/8	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	-	-	-	-	-	-	-	-	-	-	-
3/4	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	-	-	-	-	-	-	-	-	-
7/8	-	-	S	-	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	-	-	-	-	-	-	-	-	-
1----	-	-	S	-	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	-	-	-	-	-	-	-
1-1/4	-	-	-	-	S	-	S	-	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	-	S	-	-	-	-	-
1-1/2	-	-	-	-	S	-	S	-	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	-	S	-	S	-	-	-
1-3/4	-	-	-	-	-	-	-	-	S	-	S	-	S	B	S	B	S	B	S	B	S	B	S	-	S	-	S	-	-	-
2----	-	-	-	-	-	-	-	-	S	-	S	-	S	B	S	B	S	B	S	B	S	B	S	B	S	B	S	-	-	-
2-1/4	-	-	-	-	-	-	-	-	S	-	S	-	S	-	S	-	S	B	S	B	S	B	S	B	S	B	S	-	-	-
2-1/2	-	-	-	-	-	-	-	-	S	-	S	-	S	-	S	-	S	B	S	B	S	B	S	B	S	B	S	-	-	-
2-3/4	-	-	-	-	-	-	-	-	-	-	-	-	S	-	S	-	S	-	S	B	S	B	S	B	S	B	S	-	-	-
3----	-	-	-	-	-	-	-	-	-	-	-	-	S	-	S	-	S	-	S	B	S	B	S	B	S	B	S	-	S	-
3-1/2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	S	-	S	B	S	B	S	B	S	B	S	-	S	-
4----	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	S	-	S	-	S	-	S	-	S	-	S	-
4-1/2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	S	-	S	-	S	-	S	-	S	-
5-----	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	S	-	S	-	S	-	S	-	S	-

NOTE: Screws of sizes and lengths not regularly stocked may be available on orders of sufficient quantities.

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TABLE XX. - Style numbers for heads of machine and wood screws
(For use by all Federal Agencies)

General Form of Head	Style Numbers		
	Slotted	Cross recessed	Plain
Round	1s	1c	1p
Flat (countersunk) 82°	2s	2c	2p
Flat (countersunk) 100°	3s	3c	—
Fillister	4s	4c	—
Fillister, drilled	5s	—	—
Oval (countersunk)	6s	6c	—
Truss	7s	7c	—
Binding	8s	8c	—
Pan	9s	9c	—
H exagon (external)	10s	10c	10p
Hexagon socket		11	
Plutted socket		12	
Hexagon socket, flat (countersunk)		13	

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